Multivariable calculus · Differentiation in two variables

Question bank

Try each problem before reading the worked answer.

Partial derivatives, changes along a path, and a circle defined implicitly · compiled 17 October 2026

Contents

1. 1 · Partial derivatives and directional slopes
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4. Worked answers

1 Partial derivatives and directional slopes

Q1.1 · A claim about gradients

True or false: the gradient itself is the scalar slope in every direction.

S1

2 Total change and the chain rule

Q2.1 · A missing chain-rule path

A student differentiates $z = x^2 y$ along $x = t^2$, $y = 1 + t$ and writes $z' = 2xy(2t)$. Find and repair the omission.

S2

3 Implicit curves and valid tangent formulas

Q3.1 · Circle second derivative

Find y'' on $x^2 + y^2 = 25$ at $(3, 4)$.

S3

A Worked answers

1 Partial derivatives and directional slopes

Answer · Q1.1

1. A gradient contains one component per independent coordinate.
2. Choose a unit direction and take a dot product to obtain its scalar slope.

Final answer. False. Use the directional derivative.

Check. Compare the x and y slopes in W1.1.

2 Total change and the chain rule

Answer · Q2.1

1. The student included the x path but omitted the y path.
2. Add $z_y y' = x^2$. Then substitute the path.

Final answer. $z' = 4t^3 + 5t^4$; at $t = 1$, the rate is 9.

Check. Direct substitution gives z=t^4+t^5.

3 Implicit curves and valid tangent formulas

Answer · Q3.1

1. Use the circle formula $y'' = -25/y^3$.
2. Substitute $y = 4$ after checking the point and denominator.

Final answer. $-25/64$.

Check. The negative second derivative agrees with the concave-down upper semicircle.